

Problem 1 (10 Points): Let $X \sim p_\theta(x)$, where

$$p_\theta(x) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, \quad x \in \{-1, 0, 1\} \quad \text{and} \quad \theta \in (0, 1)$$

- i) (3 Points) Does $p_\theta(x)$ belong to the exponential family?
- ii) (3 Points) Is $|X|$ a complete sufficient statistic?
- iii) (4 Points) Is X complete?

Problem 2 (10 Points): Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta)$, and $\theta > 0$ is an unknown parameter. Determine a complete sufficient statistic for θ and carefully justify your answer.

Problem 3 (20 Points): Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is an unknown parameter.

- i) (10 Points) For $n = 1$, show that $T = X$ is a minimum sufficient statistic but is not a complete sufficient statistic.
- ii) (10 Points) Show that the sufficient statistic

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is not complete for any $n \geq 2$.

Problem 4 (10 Points): Prove the following law of total variance:

$$\text{Var}[Y] = \mathbb{E}[\text{Var}[Y|Z]] + \text{Var}[\mathbb{E}[Y|Z]]$$

for any real-valued random variable Y .